# Dynamic Demand

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# Where We Are

- Every demand system so far has been **static**: a consumer compares today's prices and characteristics, buys, and the period ends. Purchase $=$ consumption.

- In many markets that equality breaks, and the consumer's choice today depends on what she *expects* tomorrow:
  - **Storable goods** with frequent sales: purchase $\neq$ consumption; inventories link periods.
  - **Durable goods** with falling prices and improving quality: *when* to buy is a choice; holdings link periods.
  - **Experience goods**: consumers **learn** about match quality from use; beliefs link periods.

- Static estimates on such data mismeasure the **long-run** price elasticities that matter for markups, mergers, and welfare. And consumer dynamics are the building block for a set of supply-side questions: inter-temporal price discrimination, durable-goods monopoly, nonlinear contracts, introductory pricing.

# Road-map

- **Part I — Toolkit.** The single-agent dynamic discrete choice (DDC) model of Rust (1987): the assumptions that make it estimable, and the curse of dimensionality.
- **Part II — Storable goods.** Hendel and Nevo (2013): sales as inter-temporal price discrimination. Hendel and Nevo (2006): household-level estimation of stockpiling, and the **inclusive-value** state-space reduction.
- **Part III — Durable goods.** Coase and Bulow (1982): why the durable-goods monopolist competes with itself. Gowrisankaran and Rysman (2012): a dynamic BLP for new durables with *aggregate* data.
- **Part IV — Learning.** Relaxing Rust's assumptions (Keane–Wolpin). Bayesian learning and the value of experimentation (Erdem–Keane 1996). Crawford and Shum (2005): learning in pharmaceutical demand.

**One idea runs through all three applications**

The full state is far too large. Each paper finds an assumption that collapses it to a **low-dimensional sufficient statistic** — total inventory, an inclusive value, a posterior mean and variance.

# The Single-Agent DDC Model

- Discrete time $t$, agent $i$, finite or infinite horizon; discount factor $\beta \in [0, 1)$.
- State $s_{it}$; discrete action $a_{it} \in A = \{0, 1, \ldots, J\}$.
- Beliefs about the state transition $F(s_{it+1} \mid s_{it}, a_{it})$. Almost always **rational expectations**: beliefs equal the true law of motion, which the econometrician estimates from data. This is a key identifying assumption, not an innocuous one.
- Bellman equation:

$$V(s_{it}) = \max_{a \in A} \left\{ u(a, s_{it}) + \beta \int V(s_{it+1}) \, dF(s_{it+1} \mid s_{it}, a) \right\}$$

- **Choice-specific value function**:

$$v(a, s_{it}) = u(a, s_{it}) + \beta \int V(s_{it+1}) \, dF(s_{it+1} \mid s_{it}, a)$$

  so the optimal policy is $\alpha(s_{it}) = \arg\max_{a \in A} v(a, s_{it})$.

- **Data**: a panel $(i, t)$ of actions $a_{it}$ and observed states $x_{it}$, sometimes a payoff variable $y_{it}$ (e.g., a wage). The econometric error $\epsilon_{it}$ is **structural**: it is part of the agent's state, $s_{it} = (x_{it}, \epsilon_{it})$, observed by the agent but not by us.

# Rust's Assumptions on the Unobservables

Following Aguirregabiria and Mira (2010):

- **AS (additive separability)**: $u(a, s_{it}) = u(a, x_{it}; \theta_u) + \epsilon_{it}(a)$. One unobservable per action.
- **IID**: $\epsilon_{it}$ is i.i.d. over $i$ and $t$ with distribution $G_\epsilon$.
- **CI-X (conditional independence)**: $F(x_{it+1}, \epsilon_{it+1} \mid x_{it}, \epsilon_{it}, a) = G_\epsilon(\epsilon_{it+1})\, f_x(x_{it+1} \mid x_{it}, a; \theta_f)$. The shock today does not affect the observed state tomorrow except through the action.
- **CI-Y**: an observed payoff variable does not depend on $\epsilon$: $Y(a, x_{it}, \epsilon_{it}) = Y(a, x_{it}; \theta_Y)$.
- **CLOGIT**: $\{\epsilon_{it}(a) : a = 0, 1, \ldots, J\}$ are independent type I extreme value.

**What each assumption buys**

IID and CI-X mean $\epsilon$ never has to be carried as a state: we can integrate it out once. AS and CLOGIT give closed forms for that integral. Part IV is about what happens when they fail.

# The Integrated Value Function

- Define the **integrated (ex-ante) value function** $\bar{V}(x_{it}) \equiv \int V(x_{it}, \epsilon_{it})\, dG_\epsilon(\epsilon_{it})$.
- Under AS, IID, and CI-X, integrate both sides of the Bellman equation over $\epsilon_{it}$:

$$\bar{V}(x_{it}) = \int \max_{a \in A} \left\{ u(a, x_{it}) + \epsilon_{it}(a) + \beta \sum_{x_{it+1}} \bar{V}(x_{it+1})\, f_x(x_{it+1} \mid x_{it}, a) \right\} dG_\epsilon(\epsilon_{it})$$

  This is a fixed point in $\bar{V}(\cdot)$ on the space of *observed* states only.
- The choice-specific value function splits into a deterministic part plus the shock:

$$v(a, x_{it}) + \epsilon_{it}(a), \qquad v(a, x_{it}) \equiv u(a, x_{it}) + \beta \sum_{x_{it+1}} \bar{V}(x_{it+1})\, f_x(x_{it+1} \mid x_{it}, a)$$

- So $\bar{V}(x) = \int \max_a \{v(a, x) + \epsilon(a)\}\, dG_\epsilon$ — the same object as the expected maximum utility ("social surplus") of a static random-utility model, with $v(a, x)$ playing the role of mean utility.

# CLOGIT: Two Closed Forms

- With type I extreme value shocks (recall the difference of two independent T1EV variables is logistic), the familiar static results apply to $v(a, x)$:

- **Conditional choice probabilities (CCPs)**:

$$P(a \mid x_{it}; \theta) = \frac{\exp\{v(a, x_{it})\}}{\sum_{j=0}^{J} \exp\{v(j, x_{it})\}}$$

- **Log-sum (inclusive value)** for the integral in $\bar{V}$:

$$\bar{V}(x_{it}) = \gamma_E + \log\left(\sum_{a=0}^{J} \exp\left\{u(a, x_{it}) + \beta \sum_{x_{it+1}} \bar{V}(x_{it+1})\, f_x(x_{it+1} \mid x_{it}, a)\right\}\right)$$

  $\gamma_E \approx 0.5772$ is Euler's constant; it shifts every $\bar{V}$ equally and is usually dropped.

- The integration over $\epsilon$ disappears: we iterate on a contraction in $\bar{V}$ over the grid of $x$, which is cheap when $x$ is small.

# Estimation: Nested Fixed Point

- Rust's **NFXP**: for each trial $\theta = (\theta_u, \theta_f)$, solve the fixed point for $\bar{V}$ (inner loop), form the CCPs, and maximize the likelihood (outer loop):

$$\ell(\theta) = \sum_{i,t} \log P(a_{it} \mid x_{it}; \theta) + \sum_{i,t} \log f_x(x_{it+1} \mid x_{it}, a_{it}; \theta_f)$$

- Under CI-X the likelihood separates. In practice: (1) estimate $\theta_f$ from observed transitions; (2) take it as given and estimate $\theta_u$ by NFXP.

- **Two practical facts**:
  - The discount factor $\beta$ is **not** identified nonparametrically (Magnac and Thesmar 2002); it is almost always fixed (e.g., 0.99 per month in GR).
  - The alternative is the CCP approach (Hotz and Miller 1993): estimate $P(a|x)$ directly from the data, then invert to recover $v(a, x)$ differences without solving the DP.

- The binding constraint in demand applications: the state includes **prices and characteristics of every product**. The grid for $x$ explodes — the curse of dimensionality. Each paper below is, at heart, a way around this.

# Storable Goods: Why Do Goods Go "On Sale"?

- Grocery prices sit at a regular level, punctuated by short, deep temporary price cuts. Quantity sold spikes during a sale and **dips after it**.

- Candidate theories:
  - Mixed strategies when consumers search; price discrimination between informed and uninformed consumers (Varian 1980).
  - These explain static price *dispersion*, but not the **negative serial correlation** of quantities after a sale.
  - Loss leaders to bring traffic to the store. Perhaps.

- When goods are storable, sales can be **inter-temporal price discrimination**: consumers differ in their ability or willingness to hold inventory, and a sale screens on it.

- And for demand estimation: if the spike is partly *stockpiling*, the static own-price elasticity overstates the long-run response.

# Hendel and Nevo (2013): A Two-Period Model

- Two periods, no discounting, same static demand each period, marginal cost $c$; the seller commits to $(p_1, p_2)$.
- Two consumer types:
  - **Non-storers** buy for current consumption only: $Q^{NS}(p_t)$ in each $t$.
  - **Storers** can buy in period 1 for period-2 consumption at zero storage cost. Their consumption is $Q^S(\cdot)$ at the relevant price: $p_1$ for period 1, and $\min\{p_1, p_2\}$ for period 2.
- **If** $p_1 \geq p_2$: storage is useless (you cannot store backward). Both types buy their static demands every period.
- **If** $p_1 \equiv \underline{p} < p_2$: storers buy $2Q^S(\underline{p})$ in period 1 and nothing in period 2. Non-storers are unaffected.
- So the price *sequence* screens. Profit from a "sale, then regular price" pair:

$$\pi(\underline{p}, p_{NS}) = \left(Q^{NS}(\underline{p}) + 2Q^S(\underline{p})\right)(\underline{p} - c) + Q^{NS}(p_{NS})(p_{NS} - c)$$

  The period-2 price is paid only by non-storers, so it is set at their static monopoly price $p^*_{NS}$.

# Hendel and Nevo (2013): When Sales Pay

- Benchmark: the non-discriminating monopolist solves $\max_p(Q^S(p) + Q^{NS}(p))(p - c)$ each period, with profit $\pi^*_{ND}$.
- A sale is optimal if $\pi(\underline{p}, p^*_{NS}) > 2\pi^*_{ND}$ for some $\underline{p} < p^*_{NS}$. A necessary condition is $p^*_S < p^*_{NS}$: in Robinson's terms storers are the *weak* (more elastic) market and non-storers the *strong* one. If not, a constant price is optimal.
- The sale price solves an inverse-elasticity rule for the *combined* demand $Q^{NS} + 2Q^S$ — storers get double weight, which pulls $\underline{p}$ toward $p^*_S$.
- **Longer horizons** (storage for $T$ periods): optimal prices **cycle** — a sale, then $p^*_{NS}$ until storers' inventories run out, then another sale.
- **Why this is imperfect discrimination**: non-storers also get $\underline{p}$ during the sale. Third-degree PD (if types were observed) is the upper bound.
- **Empirics** (soft drinks, store-level data): third-degree PD would raise profits 9–14% over uniform pricing; sales capture about 25–30% of that gap. Sales **raise total welfare**; non-storers lose, but storers' gains mostly offset.

# Implications for Demand Estimation

- In the two-period model, the purchase spike during the sale combines (i) storers' higher *consumption* at the lower price and (ii) their *purchase for future consumption*. Only (i) is a demand response in the usual sense.

- **Own-price elasticity**: a static regression of purchases on current price attributes both to price and overstates the long-run elasticity.

- **Cross-price elasticity**: let $j$ go on sale in period 1, so purchases of rival $-j$ fall. In period 2, $p_j$ bounces back but purchases of $-j$ *do not rise* — storers are consuming their inventory of $j$. The static model sees little substitution to $-j$, so cross-elasticities are **understated**.

- **Outside option**: post-sale periods look like consumers switched to "no purchase". Static models overstate substitution to the outside good.

- Consequence for policy: understated cross-elasticities and overstated own-elasticities both push estimated markups and merger effects **down**.

# Hendel and Nevo (2006): Question and Data

- Question: separate the short-run (purchase) response to a temporary price cut from the long-run (consumption) response, and quantify the bias of static estimates.
- The long-run vs. short-run distinction requires controlling for **past prices and inventories** — neither of which a static model sees.

- **Data**: IRI scanner data on laundry detergent, June 1991 – June 1993 (104 weeks).
  - **Store level**: nine supermarkets in two sub-markets of a large mid-west city. For each brand-size, store, and week: price, quantity, and *promotions* (feature, display).
  - **Household level**: roughly 1,000 households. Every store visit; which brand-size was bought, where, and at what price.
  - The study abstracts from store choice: most households buy mostly at one store.

- Sales are frequent: defining a sale as $\geq 5\%$ below the modal price, 43% of liquid detergent volume is sold on sale. Feature and display are correlated with, but not the same as, price cuts — so their effects are separately identified.

# Hendel and Nevo (2006): Descriptive Statistics

TABLE I

SUMMARY STATISTICS OF HOUSEHOLD-LEVEL DATA[a]

| | Mean | Median | Std | Min | Max |
|---|---|---|---|---|---|
| Demographics | | | | | |
| Income (000's) | 35.4 | 30.0 | 21.2 | <10 | >75 |
| Size of household | 2.6 | 2.0 | 1.4 | 1 | 6 |
| Live in suburb | 0.53 | — | — | 0 | 1 |
| Purchase of laundry detergents | | | | | |
| Price ($) | 4.38 | 3.89 | 2.17 | 0.91 | 16.59 |
| Size (oz.) | 80.8 | 64 | 37.8 | 32 | 256 |
| Quantity | 1.07 | 1 | 0.29 | 1.00 | 4 |
| Duration (days) | 43.7 | 28 | 47.3 | 1 | 300 |
| Number of brands bought over the 2 years | 4.1 | 3 | 2.7 | 1 | 15 |
| Brand HHI | 0.53 | 0.47 | 0.28 | 0.10 | 1.00 |
| Store visits | | | | | |
| Number of stores visited over the 2 years | 2.38 | 2 | 1.02 | 1 | 5 |
| Store HHI | 0.77 | 0.82 | 0.21 | 0.27 | 1.00 |

[a]For Demographics, Store visits, Number of brands, and Brand HHI, an observation is a household. For all other statistics, an observation is a purchase instance. Brand HHI is the sum of the square of the volume share of the brands bought by each household. Similarly, Store HHI is the sum of the square of the expenditure share spent in each store by each household.

The median household buys one container every four weeks, buys three brands over two years (brand HHI $\approx 0.5$), and shops mostly at one store. Reduced-form evidence: during sales households buy at *higher* inventory (shorter duration since last purchase) and buy larger sizes.

# Hendel and Nevo (2006): The Model

- Brands $j$ (about a dozen; top 11 $\approx 90\%$ of volume); container sizes $x$ (32–128oz in estimation); promotions $a_{jxt}$. A purchase is a choice of one container ($> 97\%$ of purchases are single units). $d_{jxt} = 1$ if household buys brand $j$ size $x$; $x = 0$ is no purchase.
- Household $h$ chooses purchases *and* consumption $c_{ht}$ to maximize

$$\sum_t \delta^{t-1} E\Big\{ u(c_{ht} + \nu_{ht}; \theta_h) - C(i_{ht+1}; \theta_h) + \sum_{j,x} d_{hjxt}\left(\alpha_h p_{jxt} + \xi_{hjx} + \beta_h a_{jxt} + \epsilon_{hjxt}\right) \Big| s_t \Big\}$$

  subject to $i_{ht+1} = i_{ht} + x_{ht} - c_{ht}$, with $c_{ht} = \sum_j c_{jht}$ and $\alpha_h < 0$.
- $\nu_{ht}$: consumption-need shock; $C(\cdot)$: storage cost; $\xi_{hjx}$: brand-size taste; $\epsilon_{hjxt}$: logit shock. Here $\delta$ is the discount factor, $\beta$ the promotion effect.
- The key modeling choice: utility from consumption depends only on *total* quantity $c_{ht}$, not on the brand consumed. Brand differentiation happens **at the moment of purchase**. So only total inventory $i_{ht}$, not the vector of brand inventories, enters the state.

# Hendel and Nevo (2006): Assumptions

- **A1**: $\nu_{ht}$ i.i.d. over time and across households.
- **A2**: prices and promotions follow an **exogenous first-order Markov** process. No supply side. A strong assumption: it is hard to write down a supply model whose equilibrium prices are first-order Markov. But it is a reasonable model of what consumers *remember*.
- **A3**: $\epsilon_{hjxt}$ i.i.d. type I extreme value (can be generalized to GEV, provided it is separable across sizes).

- **When is "differentiation only at purchase" justified?** Let consumption utility be $\sum_j R_j c_j$ plus a common term. Then $\xi_{jx}$ captures the *expected* future brand-specific utility of the $x$ units bought. This requires:
  1. brand-specific utility **linear** in consumption (no interactions across brands in consumption), and
  2. **negligible discounting** over the inter-purchase cycle (weeks), so the timing of consumption does not matter.

- State: $s_t = (i_t, \nu_t, \epsilon_t, p_t)$ where $p_t$ collects prices *and* promotions of every brand-size — still far too large.

# Hendel and Nevo (2006): The Likelihood

- **Problem 1: inventory is never observed.** Neither initial inventory nor consumption is in the data.
- Solution: use the model. Given parameters, an initial inventory, and draws of $\nu$, the optimal consumption rule generates the inventory path implied by observed purchases:

$$Pr(d_1, \ldots, d_T \mid p_1, \ldots, p_T)$$
$$= \int \prod_t Pr\big(d_t \mid p_t,\ i_t(d_{t-1}, \ldots, d_1, \nu_{t-1}, \ldots, \nu_1, i_1),\ \nu_t\big)\, dF(\nu_1, \ldots, \nu_T)\, dF(i_1)$$

  The distribution of $i_1$ is generated by the model from the first few observations.

- **Problem 2: the state space.** Factor the choice of brand $j$ and size $x$:

$$Pr(d_{jt} = 1, x_t \mid p_t, i_t, \nu_t) = Pr(d_{jt} = 1 \mid p_t, x_t, i_t, \nu_t) \times Pr(x_t \mid p_t, i_t, \nu_t)$$

- In general the first factor needs the full DP. The trick is that, under the model's structure, it does not.
- Like most dynamic demand papers, the key step is **state-space reduction**. HN do it in three steps.

# Step 1: Brand Choice Given Size Is Static

- Let $M(s_t, j, x) = \max_c \{u(c + \nu_t) - C(i_{t+1}) + \delta E[V(s_{t+1}) \mid d_{jx}, c, s_t]\}$.
- Conditional on size $x$, optimal consumption and next-period inventory do not depend on the brand (brand does not enter consumption utility). So $M(s_t, j, x) = M(s_t, x)$.
- Because $\epsilon_{jxt}$ is i.i.d. *across brands within a size*, the common term $M(s_t, x)$ cancels in the logit ratio:

$$Pr(d_{jt} = 1 \mid x_t, p_t, i_t, \nu_t) = \frac{\exp(\alpha p_{jxt} + \xi_{jx} + \beta a_{jxt})}{\sum_k \exp(\alpha p_{kxt} + \xi_{kx} + \beta a_{kxt})}$$
$$= Pr(d_{jt} = 1 \mid x_t, p_t)$$

  the sum running over brands available in size $x$ at $t$.

- So $(\alpha, \beta, \xi)$ are estimated by a **static conditional logit**, restricting each household's choice set to the size it actually bought. Consistent, not fully efficient, and no DP is solved.
- Heterogeneity: price $\times$ demographics; household-brand fixed effects; brands a household never buys are dropped from its choice set.

# Step 2: Inclusive Values

**Step 2.** Summarize all brands of size $x$ by its **inclusive value** — a quality-adjusted price index:

$$\omega_{xt} = \log\left\{\sum_k \exp(\alpha p_{kxt} + \xi_{kx} + \beta a_{kxt})\right\}$$

- Computable from step 1 estimates. Household-specific, because $\xi$ varies across households.
- **A4 (inclusive value sufficiency)**: $F(\omega_{t+1} \mid s_t) = F(\omega_{t+1} \mid \omega_t)$. Instead of ten brands $\times$ four sizes of prices, track four indices. Testable: add lagged prices to the regression of $\omega_{t+1}$ on $\omega_t$.

# Step 3: The Reduced Dynamic Problem

**Step 3.** Solve and estimate the reduced problem, in which the consumer picks only a size:

$$V(i_t, \omega_t, \epsilon_t, \nu_t) = \max_{c,x} \Big\{ u(c + \nu_t) - C(i_{t+1}) + \omega_{xt} + \epsilon_{xt} + \delta E\big[ V(i_{t+1}, \omega_{t+1}, \epsilon_{t+1}, \nu_{t+1}) \,\big|\, i_t, \omega_t, c, x \big] \Big\}$$

- HN show this yields the *same* $E(V)$ and the same size probabilities as the full problem. A3 (logit within size) and A4 (IV sufficiency) are what make the proof work.
- Solved by value-function approximation with policy iteration; estimated by maximum likelihood on the sequence of sizes bought.
- What the consumer "sees" in the reduced problem is only one quality-adjusted price per size, plus her inventory and shocks: state dimension falls from (brands $\times$ sizes) prices and promotions to (number of sizes) $+\,2$.

# Hendel and Nevo (2006): Identification

- **Static parameters**: standard. Variation over time in prices and promotions identifies $\alpha$ and $\beta$; variation in shares across products identifies $\xi$.

- **Dynamic parameters**: neither inventory nor consumption is observed. What identifies utility from consumption and storage cost?
  - The data give the probability of purchase conditional on current prices *and* on past purchases (amount bought, time since last purchase).
  - If purchase probability does not depend on past behavior, consumers buy for immediate consumption — no storage.
  - If it does (e.g., less likely to buy soon after a large purchase), inventory matters, and the strength of that dependence reveals the storage cost.

- Division of labor:
  - Given storage costs, **preferences** determine the overall **level** of demand.
  - Given preferences, **storage costs** determine **inter-purchase duration** and how much buying responds to sales (under the same price process).

# Step 1 Estimates: Brand Choice

TABLE IV

FIRST STEP: BRAND CHOICE CONDITIONAL ON SIZE[a]

| | (i) | (ii) | (iii) | (iv) | (v) | (vi) | (vii) | (viii) | (ix) | (x) |
|---|---|---|---|---|---|---|---|---|---|---|
| Price | −0.51 | −1.06 | −0.49 | −0.26 | −0.27 | −0.38 | −0.38 | −0.57 | −1.41 | −0.75 |
| | (0.022) | (0.038) | (0.043) | (0.050) | (0.052) | (0.055) | (0.056) | (0.085) | (0.092) | (0.098) |
| *Suburban dummy | | | | −0.33 | −0.30 | −0.34 | −0.33 | −0.25 | −0.45 | −0.19 |
| | | | | (0.055) | (0.061) | (0.055) | (0.056) | (0.113) | (0.127) | (0.127) |
| *Nonwhite dummy | | | | −0.34 | −0.39 | −0.38 | −0.33 | −0.34 | −0.33 | −0.26 |
| | | | | (0.075) | (0.083) | (0.076) | (0.076) | (0.152) | (0.166) | (0.168) |
| Large family | | | | −0.23 | −0.13 | −0.21 | −0.22 | −0.46 | −0.38 | −0.43 |
| | | | | (0.080) | (0.107) | (0.080) | (0.082) | (0.181) | (0.192) | (0.195) |
| Feature | | | 1.06 | 1.05 | 1.08 | 0.92 | 0.93 | 1.08 | | 1.05 |
| | | | (0.095) | (0.096) | (0.097) | (0.099) | (0.100) | (0.123) | | (0.126) |
| Display | | | 1.19 | 1.17 | 1.20 | 1.14 | 1.15 | 1.55 | | 1.52 |
| | | | (0.069) | (0.070) | (0.071) | (0.071) | (0.072) | (0.093) | | (0.093) |
| Brand dummy variable | | ✓ | ✓ | ✓ | ✓ | | | | | |
| *Demographics | | | | | ✓ | | | | | |
| *Size | | | | | | ✓ | | | | |
| Brand–size dummy variable | | | | | | | ✓ | | | |
| Brand-HH dummy variable | | | | | | | | ✓ | | |
| *Size | | | | | | | | | ✓ | ✓ |

[a]Estimates of a conditional logit model. An observation is a purchase instance by a household. Options include only products of the same size as the product actually purchased. Asymptotic standard errors are shown in parentheses.

# Step 3 Estimates: Dynamic Parameters

TABLE VI

THIRD STEP: ESTIMATES OF DYNAMIC PARAMETERS[a]

| Household Type: | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| | Urban Market | | | Suburban Market | | |
| Household Size: | 1–2 | 3–4 | 5+ | 1–2 | 3–4 | 5+ |
| Cost of inventory | | | | | | |
| Linear | 9.24 | 6.49 | 21.96 | 4.24 | 4.13 | 11.75 |
| | (0.01) | (0.02) | (0.09) | (0.01) | (0.17) | (5.3) |
| Quadratic | −3.82 | 1.80 | −35.86 | −8.20 | −6.14 | −0.73 |
| | (29.8) | (1.77) | (0.19) | (0.03) | (1.69) | (1.53) |
| Utility from consumption | 1.31 | 0.75 | 0.51 | 0.08 | 0.92 | 3.80 |
| | (0.02) | (0.09) | (0.21) | (0.03) | (0.18) | (0.38) |
| Log likelihood | 365.6 | 926.8 | 1,530.1 | 1,037.1 | 543.6 | 1,086.1 |

[a] Asymptotic standard errors are shown in parentheses. Also included are size fixed effects, which are allowed to vary by household type.

Six household types (market $\times$ family size), each with its own storage cost and consumption utility; size fixed effects by type not reported.

# Highlights of the Parameter Estimates

**Static (step 1):**

- Including feature and display cuts the price coefficient roughly in half: much of the apparent price response is a *promotion* response.
- Larger families, non-white households (lower income in this sample), and suburban shoppers are more price sensitive.
- Brand preferences scale with container size (brand dummies interacted with size).

**Inclusive-value process (step 2):**

- First-order Markov, estimated jointly across sizes; own-size lag is the strongest predictor. Adding five more lags does not improve fit.

**Dynamic (step 3):**

- Suburban households (larger homes) have lower storage costs.
- At the median inventory (about 65oz), buying 128oz instead of 64oz raises storage cost by roughly \$0.25–\$0.75, depending on type. The typical quantity discount is about \$0.40: low-storage-cost types should buy big, high-cost types should not — as observed.

# Static vs. Dynamic Elasticities

TABLE VIII

AVERAGE RATIOS OF ELASTICITIES COMPUTED FROM A STATIC MODEL TO LONG-RUN ELASTICITIES COMPUTED FROM THE DYNAMIC MODEL[a]

| | | 64 oz. | | | | | | 128 oz. | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Brand | Size (oz.) | All[b] | Wisk | Surf | Cheer | Tide | Private Label | All[b] | Wisk | Surf | Cheer | Tide | Private Label |
| All[b] | 64 | 1.03 | 0.13 | 0.14 | 0.12 | 0.13 | 0.15 | 0.14 | 0.17 | 0.17 | 0.18 | 0.21 | 0.34 |
| | 128 | 0.17 | 0.24 | 0.26 | 0.20 | 0.28 | 0.35 | 1.23 | 0.09 | 0.11 | 0.09 | 0.15 | 0.22 |
| Wisk | 64 | 0.14 | 1.20 | 0.13 | 0.17 | 0.12 | 0.13 | 0.16 | 0.22 | 0.14 | 0.22 | 0.25 | 0.20 |
| | 128 | 0.25 | 0.27 | 0.23 | 0.31 | 0.26 | 0.28 | 0.08 | 1.42 | 0.08 | 0.13 | 0.18 | 0.11 |
| Surf | 64 | 0.14 | 0.13 | 0.93 | 0.16 | 0.13 | 0.14 | 0.18 | 0.18 | 0.12 | 0.18 | 0.22 | 0.28 |
| | 128 | 0.25 | 0.22 | 0.18 | 0.27 | 0.25 | 0.18 | 0.12 | 0.11 | 1.20 | 0.08 | 0.15 | 0.14 |
| Cheer | 64 | 0.12 | 0.17 | 0.16 | 0.84 | 0.09 | 0.13 | 0.14 | 0.24 | 0.16 | 0.14 | 0.22 | 0.24 |
| | 128 | 0.25 | 0.26 | 0.26 | 0.12 | 0.23 | 0.22 | 0.09 | 0.12 | 0.06 | 0.89 | 0.15 | 0.07 |
| Tide | 64 | 0.16 | 0.17 | 0.13 | 0.13 | 1.26 | 0.15 | 0.22 | 0.28 | 0.16 | 0.26 | 0.22 | 0.37 |
| | 128 | 0.25 | 0.31 | 0.22 | 0.24 | 0.22 | 0.31 | 0.11 | 0.16 | 0.08 | 0.13 | 1.44 | 0.31 |
| Solo | 64 | 0.15 | 0.12 | 0.15 | 0.14 | 0.12 | 0.14 | 0.17 | 0.15 | 0.15 | 0.30 | 0.30 | 0.28 |
| | 128 | 0.23 | 0.20 | 0.24 | 0.21 | 0.21 | 0.25 | 0.07 | 0.07 | 0.06 | 0.16 | 0.17 | 0.21 |
| Era | 64 | 0.21 | 0.12 | 0.13 | 0.13 | 0.10 | 0.19 | 0.43 | 0.17 | 0.15 | 0.22 | 0.19 | 0.35 |
| | 128 | 0.31 | 0.22 | 0.24 | 0.25 | 0.17 | 0.38 | 0.19 | 0.08 | 0.09 | 0.11 | 0.10 | 0.22 |
| Private | 64 | 0.19 | 0.15 | 0.14 | 0.17 | 0.17 | 1.02 | 0.32 | 0.22 | 0.15 | 0.26 | 0.31 | 0.25 |
| label | 128 | 0.29 | 0.28 | 0.34 | 0.30 | 0.39 | 0.29 | 0.16 | 0.12 | 0.13 | 0.10 | 0.27 | 1.29 |
| No purchase | | 2.12 | 1.13 | 1.15 | 1.40 | 1.27 | 2.39 | 1.80 | 7.60 | 2.26 | 14.11 | 2.38 | 10.86 |

[a]Cell entries $i$ and $j$, where $i$ indexes row and $j$ indexes column, give the ratio of the (short-run) elasticities computed from a static model divided by the long-run elasticities computed from the dynamic model. The elasticities for both models are the percent change in market share of brand $i$ with a 1 percent change in the price of $j$. The static model is identical to the model estimated in the first step, except that brands of all sizes are included as well as a no-purchase decision, not just products of the same size as the chosen option. The results from the dynamic model are based on the results presented in Tables IV–VI.

[b]Note that "All" is the name of a detergent produced by Unilever.

Ratio of static (short-run) to dynamic long-run elasticities. Diagonal $> 1$: static model **overstates own-price elasticities by about** 30%. Off-diagonal $\ll 1$: **understates cross-elasticities by up to a factor of 5**. Last row: **overstates substitution to no purchase by over** 200%.

# Hendel and Nevo: What to Take Away

- The three biases have the same sign for antitrust: a static model understates price-cost margins (own-elasticities too high) and understates diversion between products (cross-elasticities too low). Static estimates would under-predict merger price effects.

- **The method**: brand choice is static conditional on size; dynamics are carried by one inclusive value per size. This idea — **split the choice, summarize the state with an inclusive value** — is reused by Gowrisankaran–Rysman and much later work.

- **The costs**:
  - Brand differentiation only at purchase rules out, e.g., variety-seeking in consumption.
  - Prices are exogenous and first-order Markov — no supply side. Hendel and Nevo (2013) add a seller, but with a simpler demand model and market-level data.
  - Inventory is identified entirely through the model.

- Household panel data is essential here: purchase *histories* are what identify storage.

# Durable-Goods Monopoly: the Coase Conjecture

Coase (1972), formalized by Bulow (1982):

- A monopolist selling a durable good **cannot commit** to limit its own future production.
- After selling to the high-value consumers, it always has an incentive to sell one more unit to the remaining ones, as long as price exceeds marginal cost.
- Forward-looking consumers foresee prices falling toward MC, so they will not pay much today.
- If the seller can revise prices very quickly, it forfeits essentially **all** market power: the monopolist competes with its future self.

- **Renting** rather than selling solves the problem: the monopolist keeps ownership of the stock, so it gains nothing by flooding the market.
- But renting is not always feasible — e.g., renters' *moral hazard* (they do not maintain a good they do not own).

# Bulow (1982): Setup and the Rental Benchmark

- Two periods (the logic extends to any finite horizon). $MC = 0$, discount factor $\beta$. The good lasts both periods.
- Inverse demand for the *services* of the good, each period: $p = a - bq$, where $q$ is the **cumulative** stock in use.
- **Competitive benchmark**: the whole efficient stock is produced at once, $q_{1C} = a/b$, $q_{2C} = 0$.
- **Renting monopolist**: chooses the stock in each period, $q_{1R}$ and additions $q_{2R}$:

$$\max_{q_{1R}, q_{2R}} \; q_{1R}(a - bq_{1R}) + \beta(q_{1R} + q_{2R})\big[a - b(q_{1R} + q_{2R})\big]$$

  Each period is a static monopoly in the stock: $q_{1R} = a/(2b)$, $q_{2R} = 0$.
- Rental profit $= (1 + \beta)\,\dfrac{a^2}{4b}$ — the static monopoly profit, earned twice. The monopolist keeps its market power by *limiting the stock*.

# Bulow (1982): The Selling Monopolist

- **Period 2**: given first-period sales $q_{1S}$, the residual demand is $(a - bq_{1S}) - bq_{2S}$. The seller is a monopolist on it:

$$q_{2S} = \frac{a - bq_{1S}}{2b}, \qquad p_2(q_{1S}) = \frac{a - bq_{1S}}{2}$$

- **Period 1**: with rational expectations only high-value consumers buy early. The marginal buyer (value $a - bq_{1S}$ per period) is indifferent between buying now and waiting:

$$(1+\beta)(a - bq_{1S}) - p_1 = \beta\big(a - bq_{1S} - p_2(q_{1S})\big)$$

$$\Rightarrow\ p_1(q_{1S}) = \left(1 + \frac{\beta}{2}\right)(a - bq_{1S})$$

- The seller then solves

$$\max_{q_{1S}}\ q_{1S}\, p_1(q_{1S}) + \beta\,\frac{(a - bq_{1S})^2}{4b}$$

$$\Rightarrow\ q^*_{1S} = \frac{a/b}{2 + \beta/2}, \quad q^*_{2S} = \frac{a}{2b}\left(\frac{1 + \beta/2}{2 + \beta/2}\right)$$

- With $k \equiv 2 + \beta/2$, selling profit $= a^2(k-1)^2/(2bk)$; at $\beta = 1$, $0.45\,a^2/b$ vs. rental $0.50\,a^2/b$. Profit is **always** below renting, and output $q^*_{1S} + q^*_{2S} > a/(2b)$.

# Bulow (1982), Figure 1

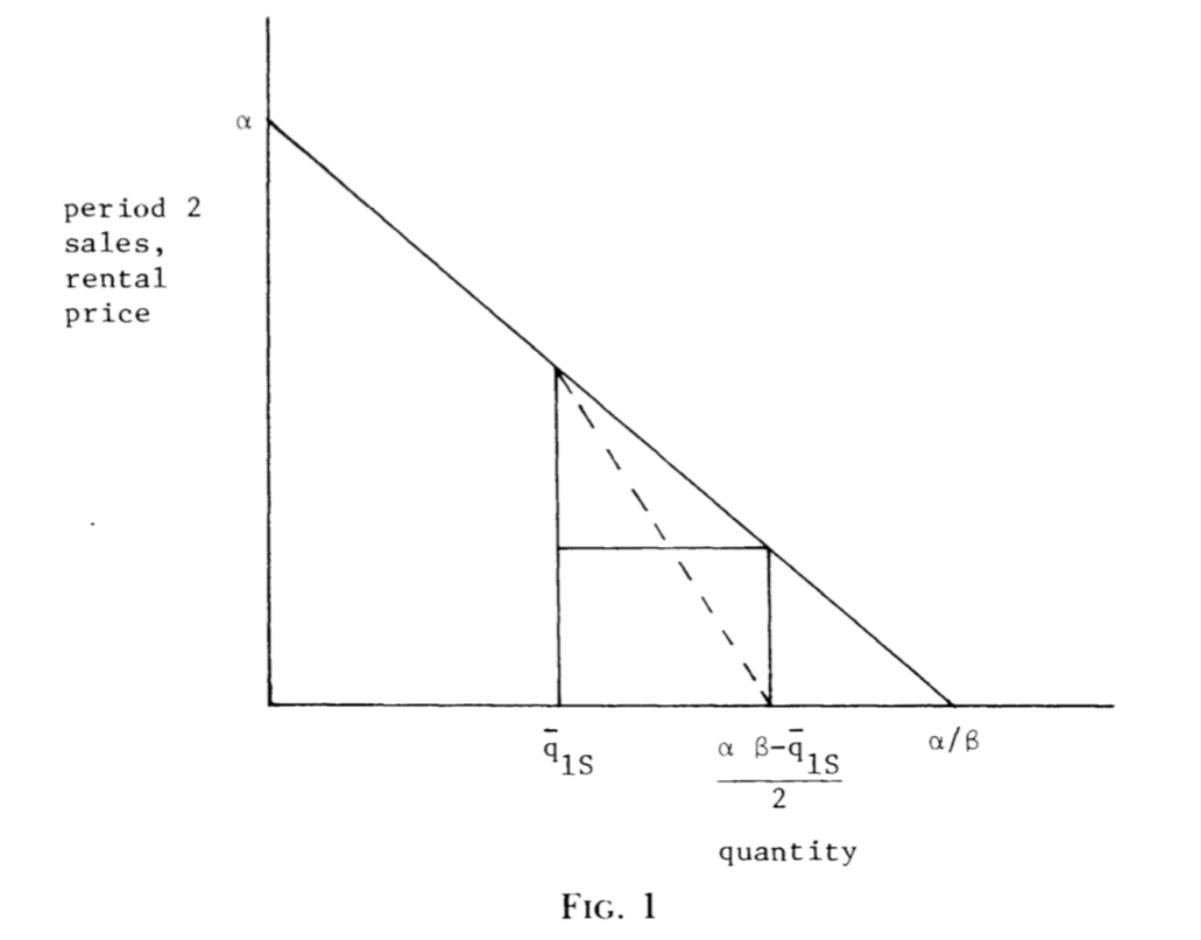



FIG. 1

Period-2 residual demand. Bulow writes demand as $\alpha - \beta q$; in our notation $\alpha \to a$, $\beta \to b$ (our $\beta$ is the discount factor). Given $\bar{q}_{1S}$, the seller adds $(a/b - \bar{q}_{1S})/2$ units in period 2.

# Durable-Goods Monopoly: Intuition and Remedies

- Why is the seller worse off, despite a more *flexible* schedule? Consumers know the seller cannot commit to keep the period-2 price high. Lower-value consumers wait, which lowers what high-value consumers will pay in period 1.
- The two-period model has the Coase logic in a milder form: consumers must wait a full period to get the lower price, so some market power survives. As the period length shrinks, profit goes to zero.
- How can the monopolist restore commitment?
  - Rent or lease instead of selling (Xerox, IBM historically).
  - Limit durability (planned obsolescence).
  - Restrict future production: capacity constraints, "limited editions", destroying molds.
  - Most-favored-customer clauses or buy-back guarantees.
  - Introduce new versions that make the old stock obsolete.
- For the empirics: when buyers are forward-looking, *when* to buy is part of demand, and today's sales depend on expected future prices. That is the subject of Gowrisankaran and Rysman.

# Gowrisankaran and Rysman (2012): Motivation

- New consumer durables: prices fall fast and quality improves fast. Consumers choose **what** to buy *and* **when** to buy, and may **upgrade** later.

- Ignoring dynamics biases price elasticities and welfare measures:
  - A consumer who waits is responding to $p_t - \beta p_{t+1}$ (the "rental cost"), not to $p_t$.
  - High-value consumers buy first, so the pool of remaining buyers changes over time.

- **Data**: monthly market-level data on digital camcorders, 2000–2006: units sold, average price $p_{jt}$, characteristics $x_{jt}$ (size, pixels, zoom, LCD size, media). 343 models, 4,436 model-months.
  - Average price falls from about \$1,000 to under \$400 (Jan. 2000 dollars), while pixel counts double and size halves.
  - The number of models grows from under 30 to almost 100.

# Gowrisankaran and Rysman (2012): vs. Hendel and Nevo

The paper differs from Hendel and Nevo (2006) on several technical dimensions:

- **Market-level data** rather than household panels. Identification must come from aggregate shares, as in BLP.
- **The current purchase enters future utility directly**: what you buy today is what you hold tomorrow, so brand choice is *not* static. In HN, the purchase affects the future only through the size (inventory).
- **Persistent heterogeneity** in preferences (BLP random coefficients). Each consumer type has its own value function and its own expectations.
- **Endogenous prices** handled with BLP instruments.
- The last two features add substantial computational burden: one DP per simulated consumer, inside the BLP contraction.

# Gowrisankaran and Rysman (2012): The Model

- Consumer $i$ in month $t$ chooses one of $J_t$ products or nothing. On upgrading, she scraps the old product: no utility from it afterwards (no resale market).
- Flow utility of product $j$, BLP-style: $\delta_{ijt}^f = \alpha_i^x x_{jt} + \xi_{jt}$. Utility in the period of purchase:

$$u_{ijt} = \delta_{ijt}^f - \alpha_i^p \ln(p_{jt}) + \epsilon_{ijt}$$

- No purchase: $u_{i0t} = \delta_{i0t}^f + \epsilon_{i0t}$, where $\delta_{i0t}^f$ is the flow utility of the product she currently owns (0 if none). She only needs to remember this scalar, not the product's identity.
- A buyer of $j$ at $t$ receives $\delta_{ijt}^f$ every subsequent period until she next buys.
- $\Omega_t$: industry state (all current $(x_{jt}, \xi_{jt}, p_{jt})$ and anything that predicts their evolution); $\epsilon_{it} = (\epsilon_{i0t}, \ldots, \epsilon_{iJ_t t})$.
- **Assumptions**: $\Omega_t$ is first-order Markov, $P(\Omega_{t+1}|\Omega_t)$, consistent with firms' optimal pricing; a continuum of consumers so no individual affects $\Omega$; $\epsilon_{it}$ i.i.d. T1EV.
- $(\alpha_i^x, \alpha_i^p)$ has mean $\alpha$ and (diagonal) variance $\Sigma$.

# The Consumer's Problem and the Inclusive Value

- Integrated value function, with $\bar{V}_i(\delta^f, \Omega) = \int V_i(\epsilon, \delta^f, \Omega)\, dP_\epsilon$:

$$\bar{V}_i(\delta^f_{i0t}, \Omega_t) = \int \max \Big\{ \delta^f_{i0t} + \beta E[\bar{V}_i(\delta^f_{i0t}, \Omega_{t+1})|\Omega_t] + \epsilon_{i0t}, \max_{j=1,\ldots,J_t} \big\{ \delta^f_{ijt} - \alpha^p_i \ln(p_{jt}) + \beta E[\bar{V}_i(\delta^f_{ijt}, \Omega_{t+1})|\Omega_t] + \epsilon_{ijt} \big\} \Big\}\, dP_\epsilon$$

- Same curse of dimensionality as HN: $\Omega_t$ is huge. Define the value of buying $j$ and the **logit inclusive value** of buying anything:

$$\delta_{ijt}(\Omega_t) = \delta^f_{ijt} - \alpha^p_i \ln(p_{jt}) + \beta E[\bar{V}_i(\delta^f_{ijt}, \Omega_{t+1})|\Omega_t]$$

$$\delta_{it}(\Omega_t) = \ln\left( \sum_j \exp(\delta_{ijt}(\Omega_t)) \right)$$

- $\delta_{it}$ is the ex-ante present discounted value of buying the preferred product today.
- **Compare with HN**: the buy/no-buy decision (size in HN) depends on the inclusive value only. But the choice of $j$ is **not** static: $j$ determines all future flow utility, so $E[\bar{V}_i(\delta^f_{ijt}, \cdot)]$ differs across $j$.

# Inclusive Value Sufficiency and the Simplified Problem

- **Assumption (IVS)**: if $\delta_{it}(\Omega_t) = \delta_{it}(\Omega_t')$, then $P(\delta_{it+1}|\Omega_t) = P(\delta_{it+1}|\Omega_t')$. Consumers forecast the future using only today's inclusive value.
- Then the state collapses to two scalars, $(\delta_{i0t}^f, \delta_{it})$:

$$\bar{V}_i(\delta_{i0t}^f, \delta_{it}) = \ln\left(\exp(\delta_{it}) + \exp\left(\delta_{i0t}^f + \beta E[\bar{V}_i(\delta_{i0t}^f, \delta_{it+1}) \mid \delta_{it}]\right)\right)$$

- Expectations are a simple AR(1), with consumer-specific coefficients:

$$\delta_{it+1} = \gamma_{1i} + \gamma_{2i}\delta_{it} + \nu_{it+1}$$

  With $0 < \gamma_{2i} < 1$, consumers expect $\delta$ to converge to $\gamma_{1i}/(1-\gamma_{2i})$.
- *Iterating among the AR(1), the Bellman equation, and the definition of $\delta_{it}$ solves the consumer problem* — a fixed point, because $\gamma$ must be consistent with the $\delta$'s the model generates.
- **Purchase probability** = Pr(buy anything) $\times$ Pr($j$ | buy):

$$\hat{s}_{ijt}(\delta_{i0t}^f, \delta_{ijt}, \delta_{it}) = \frac{\exp(\delta_{it})}{\exp\left(\bar{V}_i(\delta_{i0t}^f, \delta_{it})\right)} \times \frac{\exp(\delta_{ijt})}{\exp(\delta_{it})}$$

# Estimation: Three Nested Loops

Parameters: $\alpha = (\alpha^x, \alpha^p)$ and $\Sigma$. The discount factor is fixed at $\beta = 0.99$ (monthly) — waiting can be explained by patience or by low valuation, which are hard to tell apart.

- **Inner loop** (per simulated consumer): draw $\alpha_i = \alpha + \Sigma^{1/2}\tilde{\alpha}_i$; solve the simplified DP and the IVS fixed point. Starting at $t = 0$ with everyone holding nothing, use $\hat{s}_{ijt}$ to update the **distribution of holdings** $\delta^f_{i0t}$ period by period.

- **Middle loop**: average across consumers to get model shares $\hat{s}_{jt}(\bar{\delta}^f, \alpha^p, \Sigma)$. Invert for the mean flow utility with a BLP-type iteration

  $$\bar{\delta}^{f,\text{new}}_{jt} = \bar{\delta}^{f,\text{old}}_{jt} + \psi\big(\ln s_{jt} - \ln \hat{s}_{jt}(\bar{\delta}^{f,\text{old}})\big), \quad \psi = 1 - \beta,$$

  where $\delta^f_{ijt} = \bar{\delta}^f_{jt} + (\alpha^x_i - \alpha^x)x_{jt}$. ($\psi$ dampens the update; this is not a proven contraction.)

- **Outer loop**: $\xi_{jt} = \bar{\delta}^f_{jt} - \alpha^x x_{jt}$; minimize the GMM criterion $G(\alpha, \Sigma)' W G(\alpha, \Sigma)$ with $G = z'\xi(\alpha, \Sigma)$.

- Dubé, Fox, and Su (2012) recast this as a **mathematical program with equilibrium constraints** (MPEC): the inner fixed points become constraints, which avoids nested tolerances and speeds estimation.

# Gowrisankaran and Rysman: Identification

- The parameters are the same as in a static BLP model — the mean and variance of tastes for characteristics. What changes is the mapping from tastes to shares.
- Given the dynamic model, the comovement of characteristics, prices, and shares — across products within a month and over time — identifies $(\alpha, \Sigma)$.
- Price endogeneity: standard BLP instruments — own characteristics, mean characteristics of all products, counts of products by the same firm and by all firms. Intuition: they shift markups or substitutability, not $\xi_{jt}$.
- **Repeat purchases** are hard to identify from aggregate sales alone (a sale to a first-time buyer and an upgrade look the same). One specification adds a **micro-moment** (Petrin 2002): the change in household penetration from survey data.

# GR (2012): Number of Brands and Models

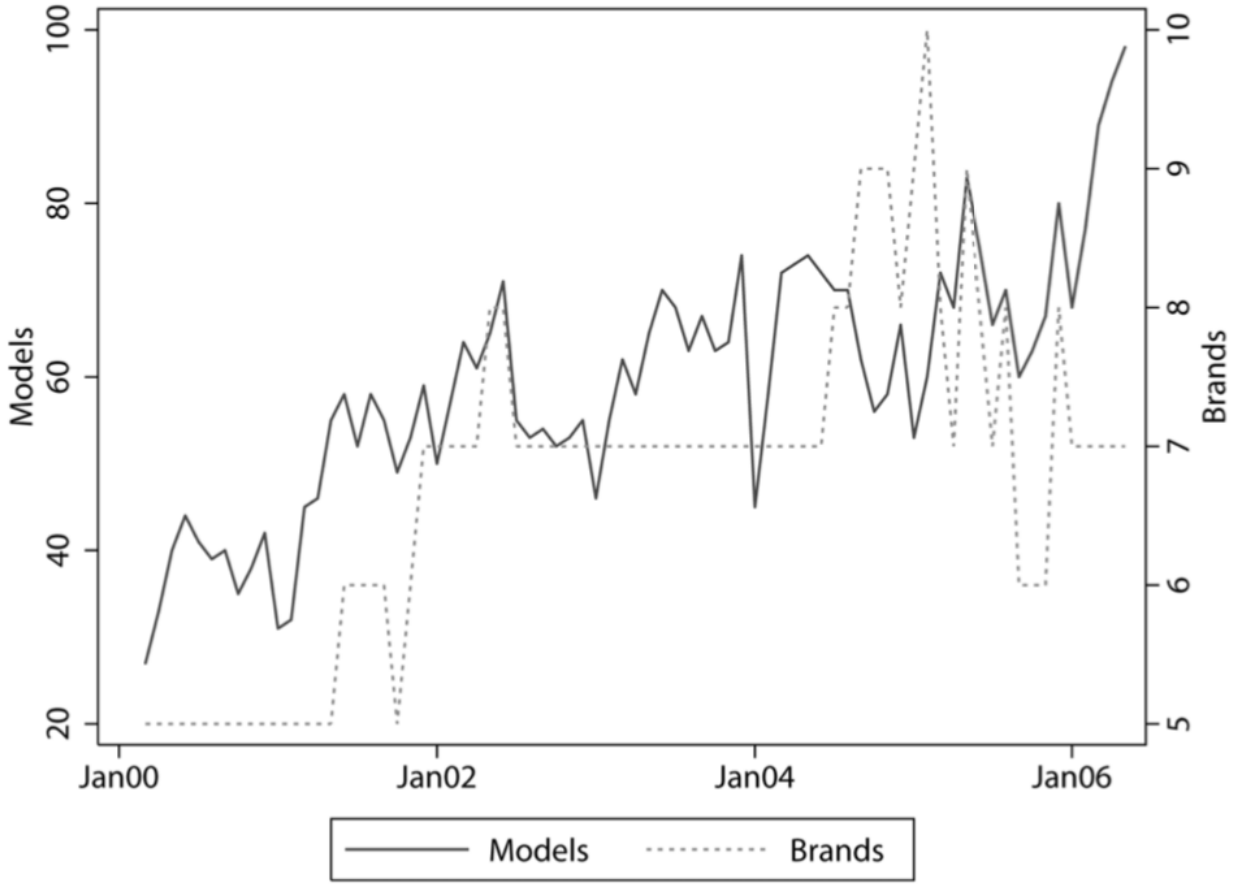



FIG. 1.—Number of brands and models over time

# GR (2012): Prices and Sales

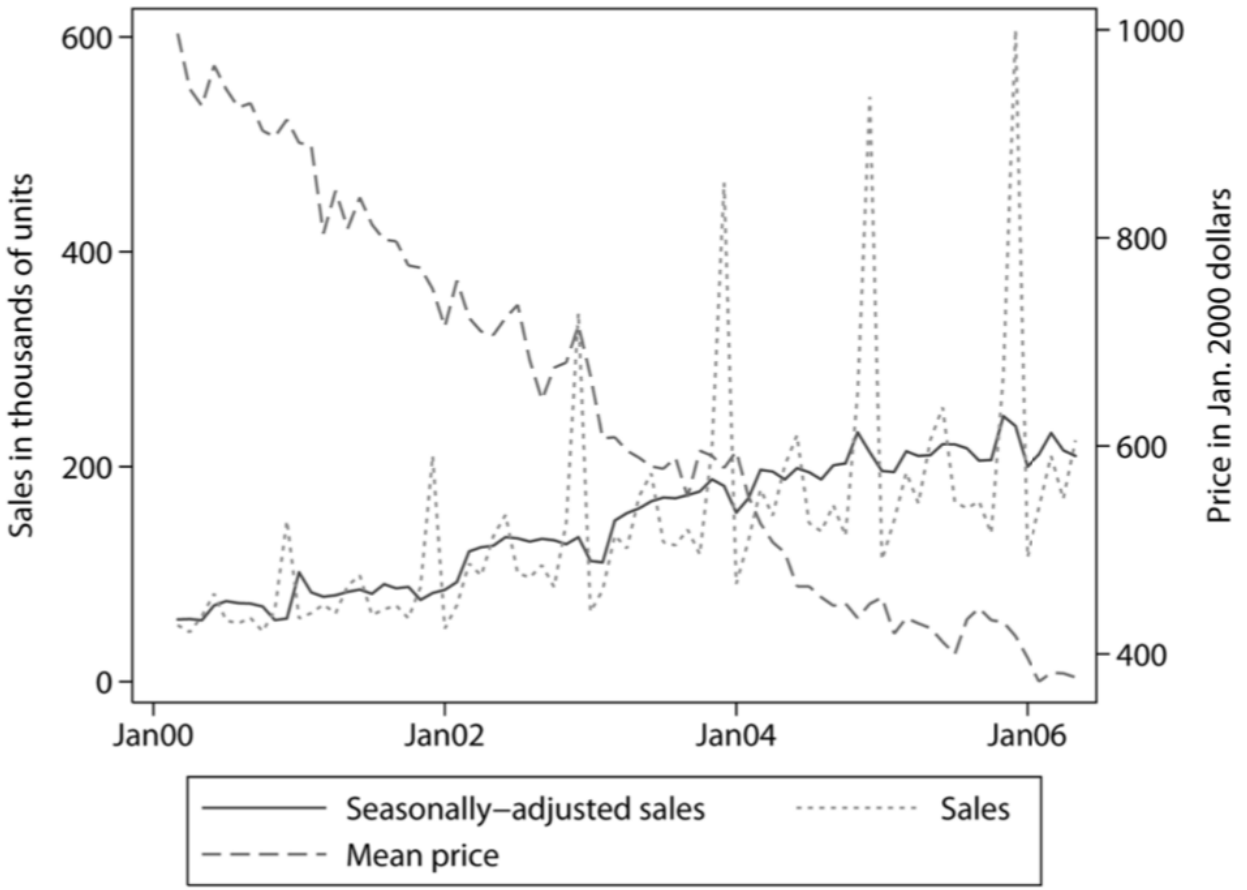



FIG. 2.—Prices and sales for camcorders

# GR (2012): Quality Improvement

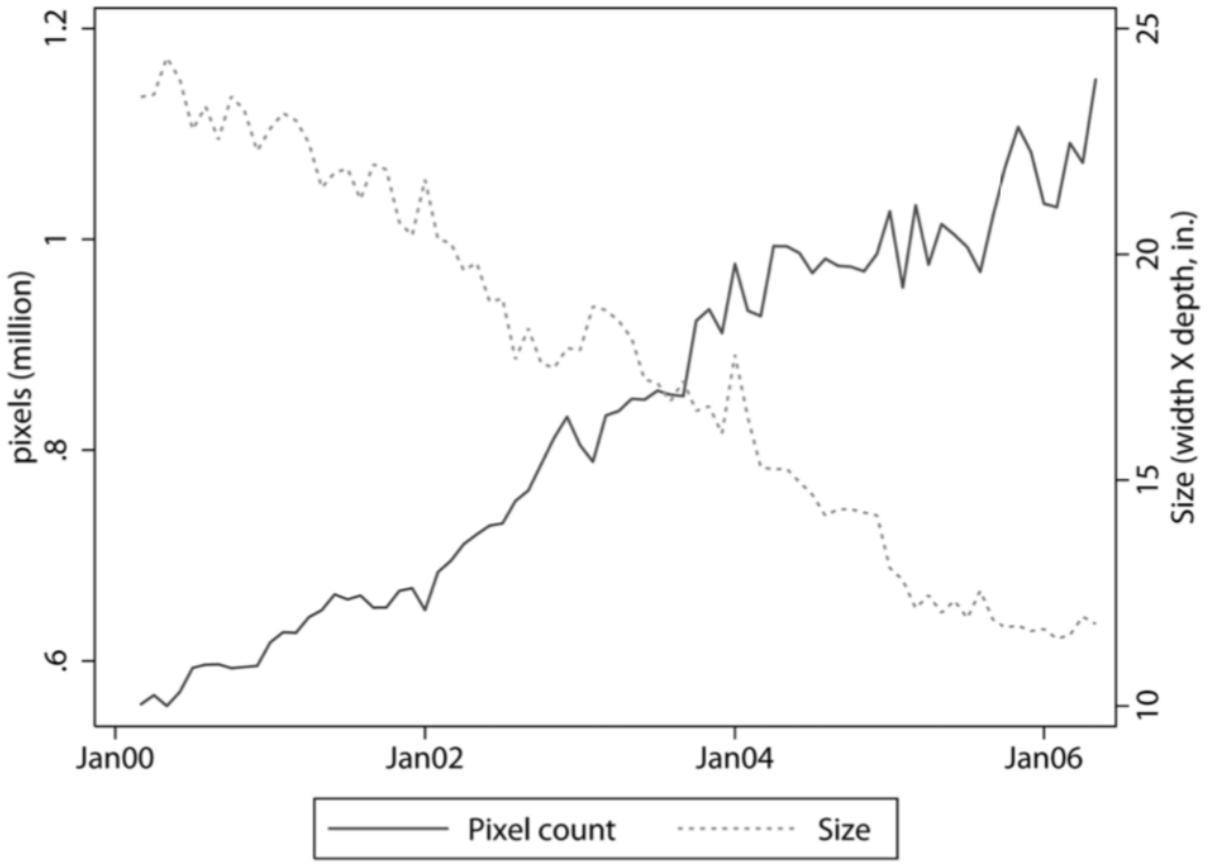



FIG. 3.—Pixel count and camcorder size over time

# GR (2012): Parameter Estimates

TABLE 1
PARAMETER ESTIMATES

| Parameter | Base Dynamic Model (1) | Dynamic Model without Repurchases (2) | Static Model (3) | Dynamic Model with Micro Moment (4) |
|---|---|---|---|---|
| Mean coefficients ($\alpha$): | | | | |
| Constant | −.092 (.029)* | −.093 (7.24) | −6.86 (358) | −.367 (.065)* |
| Log price | −3.30 (1.03)* | −.543 (3.09) | −.099 (148) | −3.43 (.225)* |
| Log size | −.007 (.001)* | −.002 (.116) | −.159 (.051)* | −.021 (.003)* |
| (Log pixel)/10 | .010 (.003)* | −.002 (.441) | −.329 (.053)* | .027 (.003)* |
| Log zoom | .005 (.002)* | .006 (.104) | .608 (.075)* | .018 (.004)* |
| Log LCD size | .003 (.002)* | .000 (.141) | −.073 (.093) | .004 (.005) |
| Media: | | | | |
| DVD | .033 (.006)* | .004 (1.16) | .074 (.332) | .060 (.019)* |
| Tape | .012 (.005)* | −.005 (.683) | −.667 (.318)* | .015 (.018) |
| Hard drive | .036 (.009)* | −.002 (1.55) | −.647 (.420) | .057 (.022)* |
| Lamp | .005 (.002)* | −.001 (.229) | −.219 (.061)* | .002 (.003) |
| Night shot | .003 (.001)* | .004 (.074) | .430 (.060)* | .015 (.004)* |
| Photo capable | −.007 (.002)* | −.002 (.143) | −.171 (.173) | −.010 (.006) |
| Standard deviation coefficients ($\Sigma^{1/2}$): | | | | |
| Constant | .079 (.021)* | .038 (1.06) | .001 (1,147) | .087 (.038)* |
| Log price | .345 (.115)* | .001 (1.94) | −.001 (427) | .820 (.084)* |

NOTE.—Standard errors are in parentheses. All models include brand dummies, with Sony excluded. There are 4,436 observations.

* Statistically significant at the 5 percent level.

# Highlights of the Estimates

- The **static** BLP model gives implausible coefficients: an **insignificant price coefficient** ($-0.099$, s.e. 148), negative coefficients on pixels and LCD size. The dynamic model gives $-3.30$ (s.e. 1.03) and sensible signs. Why:
  - For a durable, $p_t - \beta p_{t+1}$ matters. Sales jump most when prices *stop* falling; a static model relating sales to $p_t$ biases the price coefficient toward zero.
  - Heterogeneity: high-value consumers buy first, so the *population* response to quality improvements shrinks over time even if each individual's response does not.
  - Within a month consumers buy the high-price, high-quality model; across months they wait for low prices. A static model cannot reconcile the two.

- **New buyers or repeat buyers?** Adding the penetration micro-moment yields more heterogeneity in price sensitivity (s.d. $0.345 \to 0.820$) and higher valuations of quality.

- **Elasticities**: for a temporary industry-wide price increase, the one-year elasticity is about 25% smaller than the one-month elasticity — consumers delay rather than forgo. For a single model the two are close: consumers switch models rather than wait.

# Gowrisankaran and Rysman: What to Take Away

- **Cost-of-living indices**: standard indices see prices falling and quantities rising and record large welfare gains. But later buyers have lower valuations (or already own the good). The dynamic model finds **smaller welfare gains in later periods** — the "new buyer" problem.

- The paper models only the **consumer side**. Prices are whatever firms set in an (unmodeled) Markov perfect equilibrium; IVS stands in for firms' behavior.
  - Counterfactuals that change pricing (a merger, a durable-goods monopolist's commitment) need a supply side. We return to it with the dynamic oligopoly tools.

- **IVS is the key and the weak point**: it is a *bounded-rationality* assumption. Consumers ignore information in $\Omega_t$ beyond $\delta_{it}$ (e.g., the announced entry of a new model). It is testable by adding elements of $\Omega_t$ to the AR(1).

- Related: Melnikov (2013) introduced the inclusive-value approach for durables (printers), without random coefficients or repeat purchase.

# Relaxing Rust: Keane and Wolpin (1997)

Career decisions of young men, as presented in Aguirregabiria and Mira (2010):

- Each year from age 16 to $T$: $a_{it} \in \{0, 1, 2, 3, 4\}$ = home, white collar, blue collar, military, school.
- Utility:
  - $u(0, s_{it}) = \omega_i(0) + \epsilon_{it}(0)$
  - $u(4, s_{it}) = \omega_i(4) - \theta_{tc1} I(h_{it} \geq 12) - \theta_{tc2} I(h_{it} \geq 16) + \epsilon_{it}(4)$ (tuition for college, graduate school)
  - $u(a, s_{it}) = W_{it}(a)$ for $a = 1, 2, 3$, with wages

$$W_{it}(a) = r_a \exp\{\omega_i(a) + \theta_{a1} h_{it} + \theta_{a2} k_{it}(a) - \theta_{a3} k_{it}(a)^2 + \epsilon_{it}(a)\}$$

  $h$: years of schooling; $k(a)$: experience in occupation $a$; $r_a$: skill price.
- Observed state $x_{it} = (h_{it}, \text{age}_{it}, k_{it}(a) : a = 1, 2, 3)$; observed payoff $y_{it} = W_{it}(a_{it})$ if working, 0 otherwise. Unobserved: $\omega_i(a) + \epsilon_{it}(a)$, with $\epsilon_{it}$ jointly normal.
- **Check Rust's assumptions**: AS fails for $a = 1, 2, 3$ ($\epsilon$ inside the exponential). IID fails ($\omega_i(a)$ is permanent). CI-Y fails (wages depend on $\epsilon$). CLOGIT fails (correlated normal shocks).

# Keane–Wolpin: Finite Mixtures and Simulation

- Model $\omega_i$ as a **finite mixture** of permanent types $\bar{\omega}^l$, $l = 1, \ldots, L$, with probabilities $\pi_l$.
  - Conditional on type, treat $\bar{\omega}^l$ as observed: IID and CI-X are restored.
  - Integrated value function of type $l$:

$$\bar{V}_l(x_{it}) = \int \max_{a \in A} \left\{ u(a, x_{it}, \bar{\omega}^l, \epsilon_{it}) + \beta \sum_{x_{it+1}} \bar{V}_l(x_{it+1})\, f_x(x_{it+1} \mid a, x_{it}) \right\} dG_\epsilon(\epsilon_{it})$$

  - The likelihood is a $\pi$-weighted mixture over types of each person's whole history.

- Without CLOGIT, the max inside $\bar{V}_l$ is a $J$-dimensional integral with no closed form. With a large $x$ space, solving the DP exactly is infeasible.

- **Keane and Wolpin (1994)**: simulation and interpolation.
  1. Compute the expected maximum by Monte Carlo at a *subset* of state points.
  2. Regress those values on a flexible function of the state.
  3. Use the fitted regression to interpolate $\bar{V}$ at all other points.

  We use exactly this method in Crawford and Shum below.

# Bayesian Learning with Normal Priors

- Prior $\theta \sim N(\mu_0, V_0)$; signal $x \,|\, \theta \sim N(\theta, \sigma^2)$. Bayes rule: $P(\theta|x) \propto P(x|\theta)\, P(\theta)$.
- The normal is a **conjugate prior**: the posterior is also normal.

$$
\begin{aligned}
P(x|\theta)\, P(\theta) &\propto \exp\Big(-\frac{(x-\theta)^2}{2\sigma^2}\Big) \exp\Big(-\frac{(\theta-\mu_0)^2}{2V_0}\Big) \\
&\propto \exp\Big(-\frac{1}{2}\Big[\Big(\frac{1}{\sigma^2}+\frac{1}{V_0}\Big)\theta^2 - 2\theta\Big(\frac{x}{\sigma^2}+\frac{\mu_0}{V_0}\Big)\Big]\Big) \\
&\propto \exp\Big(-\frac{(\theta-\mu_1)^2}{2V_1}\Big)
\end{aligned}
$$

- Posterior precision adds; posterior mean is a precision-weighted average of signal and prior:

$$
\frac{1}{V_1} = \frac{1}{\sigma^2} + \frac{1}{V_0}, \qquad \mu_1 = \frac{x/\sigma^2 + \mu_0/V_0}{1/\sigma^2 + 1/V_0} = \frac{\sigma^2\mu_0 + V_0 x}{\sigma^2 + V_0}
$$

- After $n$ signals: $\dfrac{1}{V_n} = \dfrac{n}{\sigma^2} + \dfrac{1}{V_0}$ and $\mu_n = \dfrac{x_n/\sigma^2 + \mu_{n-1}/V_{n-1}}{1/\sigma^2 + 1/V_{n-1}}$.
- Why this matters for DP: beliefs are summarized by $(\mu_n, n)$ — a **Markov** state of fixed dimension. Without normality, the whole posterior distribution would be the state.

# Erdem and Keane (1996): The Value of Trial

Learning creates an incentive to buy the unfamiliar product for its **information value**. Product $w$ has unknown $\theta_w$; product $o$ has known $\theta_o$. Information set $I_t$.

- Value of each choice:

$$V^w(I_t) = E[U(x_t^w)|I_t] + \beta\, EV(I_{t+1}|I_t, w), \qquad I_{t+1} = \{\mu_{t+1}^w, V_{t+1}^w\}$$
$$V^o(I_t) = E[U(x_t^o)|I_t] + \beta\, EV(I_{t+1}|I_t, o), \qquad I_{t+1} = I_t$$

- The **gain from trial** $G_t = \beta\big[EV(I_{t+1}|I_t, w) - EV(I_{t+1}|I_t, o)\big] \geq 0$: future choices can only improve with more information (the value function is convex in beliefs).
- Where the information comes from, with $n_t^w$ = number of past trials:

$$\mu_{t+1}^w = \frac{V_0}{V_0 n_t^w + \sigma^2} \sum_{s=0}^{t} x_s^w d_s^w + \frac{\sigma^2}{V_0 n_t^w + \sigma^2}\, \mu_0$$

  As $n_t^w \to \infty$, $\mu_{t+1}^w \to \bar{x}^w \to \theta_w$ and the posterior variance $V_t^w \to 0$: the truth is revealed.

- Two forces: **risk aversion** penalizes the uncertain product (persistence, brand loyalty); **forward-looking** consumers value the information (experimentation). Without risk aversion, it is hard to generate the observed persistence.

# Crawford and Shum (2005): Question and Data

- Each drug affects each patient differently, and patient and doctor learn the patient's idiosyncratic **match value** from experience.
- Uncertainty has two opposing effects on drug choice:
  - Risk aversion and uncertainty about effectiveness $\Rightarrow$ **persistence** in drug choice.
  - Forward-looking behavior $\Rightarrow$ incentive to **experiment** for information.
- Two dimensions of drug quality: **symptomatic** (relief, enters utility) and **curative** (raises the recovery probability, which ends treatment).

- **Data**: all anti-ulcer prescriptions to a 10% sample of patients aged 15–85 in the Rome metropolitan area, monthly, Jan. 1990 – Dec. 1992. About 310,000 prescriptions to over 55,000 patients; the estimation sample has 34,972 patients.
- Five drugs: ranitidine (Zantac), famotidine, a composite, and the PPI omeprazole, which entered at the start of the sample.
- **Institutional advantages**: insurance is universal, so there is no variation in coverage — fewer agency concerns than in the US. Prices are set by the national Drug Commission and are essentially constant over time.

# Crawford and Shum: Evidence of Learning

- On average a patient receives 2.8 prescriptions of 1.2 drugs. Very skewed: 54% receive a single prescription.

- **Switching**: a drug *spell* is a run of prescriptions of one drug.
  - Patients average 1.19 spells.
  - The average spell is 2.37 prescriptions (2.08 for spells that end in a switch): it takes time to learn whether a drug works.

- The probability of switching is driven by **experimentation** (seeking information) or by **information** (learning the current drug does not work):
  - **Early**: about 10% switch after the first prescription — consistent with experimentation.
  - **Late**: switching rises again at the end of treatment, especially for long treatments — consistent with having learned a poor match.

# Crawford and Shum (2005), Table II

TABLE II

SWITCHING PROBABILITIES OVER THE COURSE OF TREATMENT[a]

| Prescription Number | Total Treatment Length | | | | | |
|---|---|---|---|---|---|---|
| | 5 | 6 | 7 | 8 | 9 | 10 |
| 2 | 14.3 | 13.6 | 10.9 | 10.0 | 7.8 | 9.2 |
| 3 | 11.6 | 11.6 | 6.3 | 8.8 | 7.8 | 6.6 |
| 4 | 8.9 | 5.6 | 5.4 | 3.1 | 7.8 | 3.9 |
| 5 | 13.4 | 7.9 | 10.0 | 8.8 | 4.9 | 5.3 |
| 6 | | 11.3 | 6.3 | 5.7 | 2.9 | 5.3 |
| 7 | | | 9.5 | 10.0 | 7.8 | 11.8 |
| 8 | | | | 8.1 | 4.9 | 11.8 |
| 9 | | | | | 7.8 | 5.3 |
| 10 | | | | | | 11.8 |

[a]The $(i, j)$th entry is the percentage of treatment sequences of length $j$ in which a switch was observed during the $i$th ($i \leq j$) prescription.

Switching probabilities (%) by prescription number (rows) and total treatment length (columns), uncensored sequences.

# Crawford and Shum: The Model

- The doctor maximizes the patient's expected utility (perfect agent — no incentive problem). Patient $j$, drug $n = 1, \ldots, 5$, $d_{jnt} = 1$ if prescribed:

$$\max_{\{d_{jnt}\}} E \sum_{t=1}^{\infty} \beta^t \sum_n d_{jnt}\, u_{jnt}\, (1 - w_{jt-1})$$

  $w_{jt} = 1$ once the patient has recovered; recovery is absorbing with terminal utility 0.

- Two sources of patient-level uncertainty:
  - drug-specific **match values** $(\mu_{jn}, \nu_{jn})$ (symptomatic, curative), revealed through noisy signals $(x_{jnt}, y_{jnt})$ each time the drug is taken;
  - **treatment length**: the recovery probability depends on the curative signals received.
- **Diagnostic matching**: before prescribing, the doctor classifies the patient into one of $K$ types (unobserved by us); initial recovery probability $h_{0j} = \theta_k$ with prob. $p_k$. Type also shifts the priors over match values.
- **CARA preferences**:

$$u(x_{jnt}, p_n, \epsilon_{jnt}) = -\exp(-r\, x_{jnt}) - \alpha p_n + \epsilon_{jnt}, \qquad r > 0$$

  Flow utility is negative, so patients never want to delay recovery.

# Recovery and Learning

- **Recovery**: the *odds* of having recovered rise by the curative signal of the drug taken:

$$h_{jt}(h_{jt-1}, y_{jnt}) = \frac{\frac{h_{jt-1}}{1-h_{jt-1}} + d_{jnt}\, y_{jnt}}{1 + \frac{h_{jt-1}}{1-h_{jt-1}} + d_{jnt}\, y_{jnt}}, \qquad h_{j0} = \theta_k$$

- **Priors** (type-$k$ specific): $(\mu_{jn}, \nu_{jn}) \sim N\big([\underline{\mu}_{nk}, \underline{\nu}_{nk}],\, \mathrm{diag}(\underline{\sigma}_n^2, \underline{\tau}_n^2)\big)$.
- **Signals**: $(x_{jnt}, y_{jnt}) \sim N\big([\mu_{jn}, \nu_{jn}],\, \mathrm{diag}(\sigma_n^2, \tau_n^2)\big)$, independent over time.
- **Updating** (the normal formulas above), with $l_{jnt}$ the number of times $j$ has taken $n$:

$$\mu_{jn,t+1} = \frac{\sigma_n^2 \mu_{jnt} + V_{jnt} x_{jn,t+1}}{\sigma_n^2 + V_{jnt}}, \qquad V_{jn,t+1} = \frac{\underline{\sigma}_n^2 \sigma_n^2}{\sigma_n^2 + l_{jn,t+1}\underline{\sigma}_n^2}$$

  if $n$ is taken at $t+1$; unchanged otherwise. Same for $(\nu, R)$ with $(\underline{\tau}_n^2, \tau_n^2)$.
- **Rational expectations**: priors equal the true population distribution of match values. Relaxed for omeprazole (the new entrant), whose prior means may drift over five half-year periods.
- To estimate: $\underline{\mu}_{nk}, \underline{\nu}_{nk}, \underline{\sigma}_n^2, \underline{\tau}_n^2, \sigma_n^2, \tau_n^2, r, \alpha, (\theta_k, p_k)$.

# The Dynamic Drug Choice Problem

- State $S_t$: posterior means $(\mu_{jnt}, \nu_{jnt})$ and counts $I_{jnt}$ for $n = 1, \ldots, 5$; recovery probability $h_{jt-1}$; shocks $\epsilon_{jnt}$. Thanks to normality, $(\mu, I)$ summarizes beliefs.
- Bellman equation:

$$
\begin{aligned}
W(S_t) = \max_n \Big\{ & -\exp\big[-r\mu_{jnt} + \tfrac{1}{2}r^2(\sigma_n^2 + V_{jnt})\big] - \alpha p_n + \epsilon_{jnt} \\
& + \beta E\big[(1 - h_{jt}(h_{jt-1}, y_{jnt}))\, E[W(S_{t+1}) \mid x_{jnt}, y_{jnt}] \,\big|\, n, S_t\big] \Big\} \\
\equiv \ & \max_n \{W_n(S_t)\}
\end{aligned}
$$

- The first line uses the normal m.g.f.: $E[-e^{-rx}] = -\exp(-r\mu + \frac{1}{2}r^2 \cdot \text{variance})$, where the variance combines signal noise $\sigma_n^2$ and belief uncertainty $V_{jnt}$.
- $\frac{1}{2}r^2(\sigma_n^2 + V_{jnt})$ is a **risk premium**. Trying a drug lowers $V_{jnt}$, raising its expected utility next time — it acts like a **switching cost** and generates persistence.
- The outer expectation is over the signals $(x_{jnt}, y_{jnt})$ given $S_t$; the inner one applies the Bayesian updating rule.

# Solving the DP: Keane–Wolpin Simulation and Interpolation

With logit $\epsilon$, the integrated value function is a log-sum (Euler's constant dropped):

$$W(S) = \log\left\{\sum_n \exp\left(Eu_n(S) + \beta E\left[(1 - h'(h, y_n))\, E[W(S')|x_n, y_n] \,\middle|\, n, S\right]\right)\right\}$$

But $S$ is continuous and high-dimensional (16 continuous/count variables). Algorithm:

1. Draw a set of grid points $S^*$ from the state space.
2. For each $s \in S^*$, make an initial guess $W^0(s)$.
3. Fit a flexible (high-order polynomial) regression $W^0(s) = G(s)'\theta^0 + u_s$ on $S^*$.
4. For each $s$ and drug $n$, draw $M$ signals $(x_n^m, y_n^m)$ and compute the Bayesian posterior state $s^m$.
5. Approximate $E[(1 - h'(h, y_n))E[W(s')|x_n, y_n] \mid n, s] \approx \frac{1}{M}\sum_m (1 - h'(h(s), y_n^m))\, \hat{W}^0(s^m)$ using the fitted regression.
6. Update $W^1(s)$ for each $s \in S^*$ with the log-sum formula.
7. Repeat 3–6 until $W$ converges on the grid.

Estimation is by simulated maximum likelihood over each patient's drug sequence and treatment length, with right-censoring for sequences still active at the end of the sample.

# Crawford and Shum: Identification

**Key restriction**: symptomatic match values enter only utility; curative match values enter only recovery.

- **Symptomatic** $(\underline{\mu}_n, \underline{\sigma}_n^2, \sigma_n^2, r)$:
  - Initial prescription shares $\rightarrow \underline{\mu}_n$ (doctors rely on priors).
  - Shares early vs. late in treatment (late choices reflect true match values) $\rightarrow \underline{\sigma}_n^2$. If $\underline{\sigma}_n^2 = 0$, late shares would equal initial shares.
  - Overall incidence of switching $\rightarrow r$. Under rational expectations a first trial is equally likely to update up or down; only risk aversion makes the repeat probability jump after first use.
  - How switching changes with the number of prescriptions taken $I \rightarrow \sigma_n^2$ (how fast perceived variance falls).
- **Curative** $(\underline{\nu}_n, \underline{\tau}_n^2, \tau_n^2)$ and **severity** $h_{0j}$: these enter only through expected recovery probabilities. Recovery frequencies (treatment lengths) conditional on different drug sequences identify them jointly.
- **Price**: drug prices do not vary over time, so $\alpha$ and $\underline{\mu}_n$ cannot be separated with linear utility. CARA helps: $\mu_{jnt}$ enters interacted with the time-varying $V_{jnt}$.
- **Types**: add types until fit stops changing ($K = 4$). Overall: individual-level panel data is required to separate uncertainty from heterogeneity.

# Crawford and Shum (2005), Table III

TABLE III

DYNAMIC MODEL: PARAMETER ESTIMATES

| Parameter | Est. | Std. Err. | Est. | Std. Err. |
|---|---|---|---|---|
| Illness heterogeneity distribution | Recovery Probability | | Type Probability | |
| $\theta_1$ (Type 1) | 0.433 | 0.003 | 0.593 | 0.006 |
| $\theta_2$ (Type 2) | 0.127 | 0.003 | 0.335 | 0.006 |
| $\theta_3$ (Type 3) | 0.199 | 0.007 | 0.043 | 0.001 |
| $\theta_4$ (Type 4) | 0.432 | 0.011 | 0.029 | 0.002 |
| Means, symptom match values[b] | Type 1 | | Type 2 | |
| $\underline{\mu}_1$ | 0.927 | 0.282 | 1.195 | 0.369 |
| $\underline{\mu}_2$[c] | 0.928 | 0.287 | 0.428 | 0.166 |
| $\underline{\mu}_3$ | 0.481 | 0.197 | −0.028 | 0.178 |
| $\underline{\mu}_4$ | 0.335 | 0.161 | −0.145 | 0.079 |
| $\underline{\mu}_5$ | 0.451 | 0.174 | −0.483 | 0.137 |

# Crawford and Shum (2005), Table III, cont'd

| Means, curative match values[b] | Type 1 | | Type 2 | |
|---|---|---|---|---|
| $\underline{\nu}_1$ | 0.014 | 0.003 | 0.006 | 0.000 |
| $\underline{\nu}_2$[c] | 0.015 | 0.005 | 0.006 | 0.001 |
| $\underline{\nu}_3$ | 0.013 | 0.030 | 0.006 | 0.095 |
| $\underline{\nu}_4$ | 0.013 | 0.084 | 0.014 | 0.009 |
| $\underline{\nu}_5$ | −0.034 | 0.000 | −0.038 | 0.000 |
| Std. dev., symptom match values | | | | |
| $\underline{\sigma}$ | 1.574 | 0.448 | | |
| Std. devs., symptom signals | | | | |
| $\sigma_1$ | 0.998 | 0.287 | | |
| $\sigma_2$ | 1.134 | 0.326 | | |
| $\sigma_3$ | 1.375 | 0.395 | | |
| $\sigma_4$ | 1.159 | 0.333 | | |
| $\sigma_5$ | 0.931 | 0.268 | | |
| Std. dev., curative match values | | | | |
| $\underline{\tau}$ | 0.007 | 0.000 | | |
| Std. dev., curative signals | | | | |
| $\tau$ | 0.007 | 0.001 | | |
| Price coefficient, $\alpha$[a] | 1.080 | 0.091 | | |
| Risk-aversion parameter, $r$ | 0.990 | 0.274 | | |
| Discount rate, $\beta$ | 0.950 | Fixed | | |
| Number of observations | 34,972 | | | |
| Number of similar draws | 30 | | | |
| Log likelihood function | −124,484.34 | | | |

[a]Price in thousands of lire.

[b]Prior symptomatic and curative means are reported for two highest probability types. For prior means corresponding to all types, see Table IV.

[c]Prior means for drug 2 are reported for the Jul–Dec 1992 period. For prior means corresponding to earlier periods, see Table IV.

# Crawford and Shum (2005), Table IV

TABLE IV

DYNAMIC MODEL: PARAMETER ESTIMATES (CONT'D): MATCH VALUES FOR ALL TYPES AND ACROSS TIME FOR OMEPRAZOLE

| | Type 1 | | Type 2 | | Type 3 | | Type 4 | |
|---|---|---|---|---|---|---|---|---|
| Parameter | Est. | Std. Err. | Est. | Std. Err. | Est. | Std. Err. | Est. | Std. Err. |
| Match values, all types | | | | | | | | |
| Symptom match values | | | | | | | | |
| $\underline{\mu}_1$ | 0.927 | 0.282 | 1.195 | 0.369 | 0.489 | 0.163 | 0.151 | 0.091 |
| $\underline{\mu}_2$[a] | 0.928 | 0.287 | 0.428 | 0.166 | 0.577 | 0.198 | 0.573 | 0.199 |
| $\underline{\mu}_3$ | 0.481 | 0.197 | −0.028 | 0.178 | 1.762 | 0.531 | 0.013 | 0.167 |
| $\underline{\mu}_4$ | 0.335 | 0.161 | −0.145 | 0.079 | −0.111 | 0.305 | 0.504 | 0.184 |
| $\underline{\mu}_5$ | 0.451 | 0.174 | −0.483 | 0.137 | −0.113 | 0.125 | −0.561 | 0.220 |
| Curative match values | | | | | | | | |
| $\underline{\nu}_1$ | 0.014 | 0.003 | 0.006 | 0.000 | 0.011 | 0.002 | 0.014 | 0.010 |
| $\underline{\nu}_2$[a] | 0.015 | 0.005 | 0.006 | 0.001 | 0.011 | 0.006 | 0.015 | 0.003 |
| $\underline{\nu}_3$ | 0.013 | 0.030 | 0.006 | 0.095 | 0.004 | 0.001 | 0.013 | 0.329 |
| $\underline{\nu}_4$ | 0.013 | 0.084 | 0.014 | 0.009 | −0.035 | 0.214 | 0.012 | 0.003 |
| $\underline{\nu}_5$ | −0.034 | 0.000 | −0.038 | 0.000 | −0.037 | 0.054 | −0.034 | 0.409 |
| Time-varying priors for omeprazole | | | | | | | | |
| Symptom match value, $\underline{\mu}_2$ | | | | | | | | |
| Period 1 | 0.805 | 0.258 | 0.306 | 0.140 | 0.454 | 0.171 | 0.451 | 0.172 |
| Period 2 | 0.910 | 0.285 | 0.411 | 0.166 | 0.560 | 0.197 | 0.556 | 0.198 |
| Period 3 | 0.722 | 0.237 | 0.223 | 0.122 | 0.371 | 0.151 | 0.368 | 0.152 |
| Period 4 | 0.979 | 0.301 | 0.480 | 0.181 | 0.628 | 0.212 | 0.625 | 0.214 |
| Period 5[a] | 0.928 | 0.287 | 0.428 | 0.166 | 0.577 | 0.198 | 0.573 | 0.199 |
| Curative match value, $\underline{\nu}_2$ | | | | | | | | |
| Period 1 | −0.007 | 0.011 | −0.016 | 0.010 | −0.011 | 0.011 | −0.007 | 0.010 |
| Period 2 | −0.001 | 0.012 | −0.011 | 0.011 | −0.006 | 0.012 | −0.001 | 0.011 |
| Period 3 | 0.015 | 0.016 | 0.005 | 0.015 | 0.011 | 0.016 | 0.015 | 0.016 |
| Period 4 | 0.013 | 0.017 | 0.004 | 0.016 | 0.009 | 0.017 | 0.013 | 0.017 |
| Period 5[a] | 0.015 | 0.005 | 0.015 | 0.001 | 0.011 | 0.006 | 0.015 | 0.003 |

[a] Prior means for drug 2 are reported for the Jul–Dec 1992 period. Prior means corresponding to earlier periods are reported in the bottom panel of this table.

Prior mean match values by patient type, and the drift in omeprazole's prior over 1990–92.

# Crawford and Shum: Results

- **Illness heterogeneity**: Type 1 ("not so sick") patients begin with a 43% chance of recovering after the first prescription; Type 2 ("sick"), 33.5% of the sample, start at 12.7%.
- **Match-value heterogeneity is large**: the dispersion of symptomatic match values ($\underline{\sigma} = 1.58$) dwarfs differences in mean values across drugs — drugs are **horizontally** differentiated even within a type. Rankings differ between the symptomatic and curative dimensions.
- **Learning is fast**: over two-thirds of initial uncertainty is resolved after one prescription. With risk aversion this produces strong persistence after a single prescription.

- **Counterfactuals** (5,000 simulated patients):
  - *Complete information* ($V = R = 0$): discounted utility rises by about 9%; patients use more drugs ($1.4 \rightarrow 1.9$); market much less concentrated (HHI $4{,}242 \rightarrow 2{,}676$). Uncertainty-induced switching costs favor the market leader.
  - *No experimentation* (stay on first drug): utility falls about 6% below baseline. Learning gets patients most of the way to first best.
  - *No diagnostic matching*: utility falls about 11% — the doctor's initial diagnosis is at least as valuable as learning by experience.

# Three Applications, One Dimensionality Problem

| | **Hendel–Nevo (2006)**<br>laundry detergent | **Gowrisankaran–Rysman (2012)**<br>digital camcorders | **Crawford–Shum (2005)**<br>anti-ulcer drugs |
|---|---|---|---|
| Dynamic link | inventory | holdings / upgrading | beliefs (learning) |
| Data | household panel | market shares | patient panel |
| Heterogeneity | 6 household types; brand FE | random coefficients | 4 latent types |
| State reduction | total inventory; one IV per size | $(\delta^f_{i\mathbf{0}}, \delta_i)$ via IVS | $(\mu, I)$ via normal conjugacy |
| Solution | approx. policy iteration | value iteration per consumer | Keane–Wolpin interpolation |
| Estimation | 3-step MLE | BLP-GMM, 3 nested loops | simulated MLE |
| Supply side | none (exog. Markov prices) | none (IVS stands in) | none (regulated prices) |
| Headline | static: own-$\varepsilon$ +30%, cross $\div$5 | static $\hat{\alpha}^p \approx 0$; COLI overstated | fast learning; uncertainty costs $\approx$ 9% |

- The common thread: each sufficient statistic is bought with a substantive assumption — differentiation only at purchase, inclusive-value sufficiency, normal signals with rational-expectations priors.

# Additional Topics

Active areas in dynamic demand we have not covered.

- **Supply side with forward-looking consumers.** Durable-goods pricing and innovation, e.g., Nair (2007) on video games, Conlon (2012) on LCD TVs, Goettler and Gordon (2011) on CPUs. Covered with the dynamic oligopoly tools.

- **Learning and advertising.** Ackerberg (2003) separates informative from prestige effects of advertising; Ching (2010) on learning and generic entry in pharmaceuticals.

- **Expectations.** Almost every paper assumes rational expectations or a simple AR(1). Erdem, Imai, and Keane (2003) model price expectations explicitly for a storable good; identifying beliefs separately from preferences remains a central open problem.

- **Computation.** CCP estimators (Hotz–Miller 1993; Arcidiacono–Miller 2011) and MPEC (Dubé–Fox–Su 2012) make richer models feasible.

# Summary

- When purchase $\neq$ consumption (storables), when timing is a choice (durables), or when beliefs evolve (learning), **static demand estimates are biased** in predictable directions — usually toward understating market power.

- The DDC toolkit makes these models estimable, but demand applications face a huge state space. The craft is in finding the right **sufficient statistic**.

- **Three open questions worth carrying forward:**
  1. Expectations are assumed, not measured. Inclusive-value sufficiency and first-order Markov prices are convenient; how wrong are they, and does it matter for counterfactuals?
  2. The supply side is missing. All three papers take prices as exogenous. Inter-temporal price discrimination and durable-goods pricing require the firm's problem too.
  3. Discount factors are fixed, not estimated. Patience and low valuation both explain waiting; the welfare numbers depend on the choice.

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